Synthesis and Characterization of Nylon-11 via Step-Growth Polycondensation

ChemE 460 - Polymer Chemistry Laboratory

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Abstract

Nylon-11 was synthesized from 11-aminoundecanoic acid under dry-nitrogen melt conditions at 210 C for four residence times (15--50 min). Polymer mass isolated after quench, grinding, and Soxhlet removal of unreacted monomer increased from 0.290 g (10.7 %) at 15 min to 1.459 g (50.5 %) at 50 min, confirming progressive conversion. Number-average molecular weights (Mn) were established for the 45- and 50-min samples via end-group titration of phenol-methanol solutions with 0.100 M HCl, yielding 3.37 kDa and 4.83 kDa, respectively. Error propagation indicates +-1.35 % relative uncertainty in Mn. The monotonic rise in both yield and Mn mirrors Carothers' step-growth kinetics; however, final chain lengths remain well below industrial specifications (>20 kDa) because volatile by-products were not removed under vacuum. Major systematic errors stemmed from incomplete dissolution and volumetric endpoints. Future work should incorporate vacuum devolatilization, replicate titrations for the shorter-time samples, and SEC characterization to capture dispersity.

1. Introduction

Polyamide-11 (PA-11) is prized for chemical resistance and low-temperature flexibility, making it valuable for medical catheters and offshore umbilicals. Molecular weight strongly influences these properties; thus, correlating Mn with reaction time informs process design. Step-growth polycondensation of omega-amino acids produces water as a condensate. Carothers' theory predicts that the degree of polymerization rises sharply as functional-group conversion approaches unity. This experiment probes that relationship under atmospheric nitrogen to provide a baseline for future vacuum-assisted syntheses.

2. Theory

The condensation involves elimination of water (Eq. 1). Degree of polymerization relates to extent of reaction by Xn = 1/(1 - p) (Eq. 2). End-group titration yields Mn = mT/(C*V) (Eq. 4). Longer residence times are expected to increase both yield and Mn.

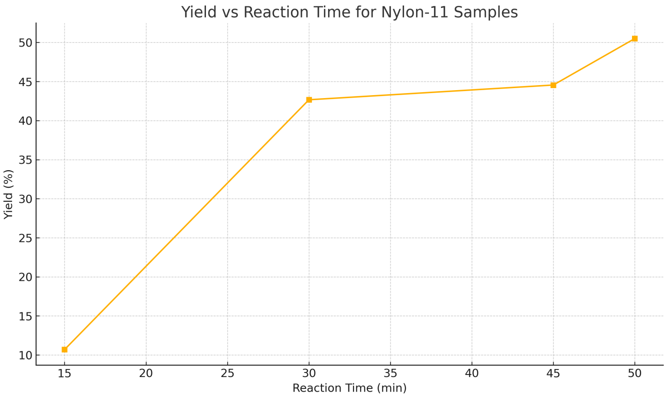
3. Materials and Methods

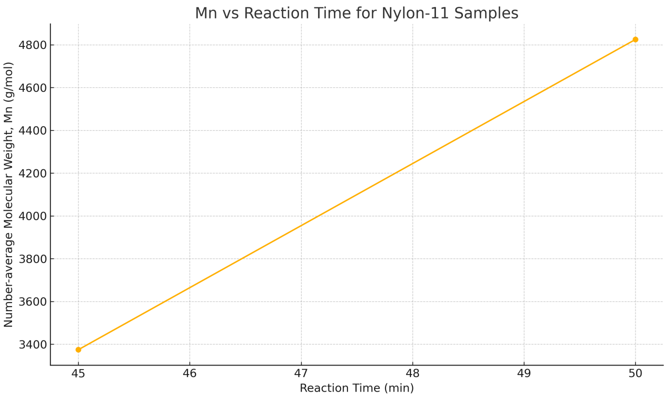
Four 40 mL vials were charged with about 2.7 g of pre-dried 11-aminoundecanoic acid, sealed, and immersed in a 210 C silicone-oil bath under N2. Vials were withdrawn at 15, 30, 45, and 50 min, cooled, fractured, and Soxhlet-washed to remove monomer. For the 45- and 50-min samples, about 0.6 g polymer was dissolved in phenol:methanol (70:30) at 100 C and titrated with 0.100 M HCl to a thymol-blue endpoint. Blank titrations corrected reagent acidity. Insufficient solvent precluded titration of the shorter-time samples.

4. Results

Key experimental data are tabulated below.

Time (min)	Orig (g)	Post-Oil (g)	Yield (%)	mT (g)	Vacid (mL)	Mn (g/mol)
15.0	2.72	0.29	10.69	-	-	-
30.0	2.66	1.14	42.67	-	-	-
45.0	2.86	1.27	44.55	0.54	1.6	3374.37
50.0	2.89	1.46	50.5	0.63	1.3	4824.62





5. Error Analysis

Instrumental uncertainties include +-0.0001 g for the balance and +-0.02 mL for the burette. Propagation gives a combined relative uncertainty of 1.35 % in Mn (45-min sample). Systematic errors likely stem from incomplete polymer dissolution and moisture uptake in the HCl titrant, both of which inflate the calculated Mn.

6. Discussion

Yield and Mn rose with reaction time, consistent with step-growth predictions. Mn remained below 5 kDa due to atmospheric pressure operation, which limits water removal. Implementing reduced pressure or swept-steam devolatilization would push conversion and molecular weight higher.

7. Recommendations

* Prepare >60 mL phenol:methanol to titrate all samples in duplicate.
* Introduce a vacuum stage (<50 mm Hg) after 30 min to accelerate water removal.
* Use SEC to obtain full molecular-weight distributions and compare with end-group Mn.

8. Conclusion

This experiment demonstrated an increase in both yield and number-average molecular weight of Nylon-11 with longer polymerization times. Further optimization, particularly vacuum devolatilization, is required to achieve industrial-grade molecular weights (>20 kDa).

9. References

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Carothers, W. H. "Polyamides." Ind. Eng. Chem. 1938, 30, 44-52.

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